



5.2.4 Gaussian elimination via Gauss transforms

05.2.4 Gauss transforms

fit width



Definition 5.2.4.1. A matrix L_k of the form

$$L_k = \left(\begin{array}{c|cc} I_k & 0 & 0 \\ \hline 0 & 1 & 0 \\ 0 & l_{21} & I \end{array} \right),$$

where I_k is the $k \times k$ identity matrix and I is an identity matrix "of appropriate size" is called a Gauss transform.

Gauss transforms, when applied to a matrix, take multiples of the row indexed with k and add these multiples to other rows. In our use of Gauss transforms to explain the LU factorization, we subtract instead:

Example 5.2.4.2. Evaluate

$$\left(\begin{array}{c|ccc} 1 & 0 & 0 & 0 \\ \hline 0 & 1 & 0 & 0 \\ 0 & -\lambda_{21} & 1 & 0 \\ 0 & -\lambda_{31} & 0 & 1 \end{array} \right) \left(\begin{array}{c} \tilde{a}_0^T \\ \tilde{a}_1^T \\ \tilde{a}_2^T \\ \tilde{a}_3^T \end{array} \right) =$$

▼ [Solution](#)

$$\left(\begin{array}{c|ccc} 1 & 0 & 0 & 0 \\ \hline 0 & 1 & 0 & 0 \\ 0 & -\lambda_{21} & 1 & 0 \\ 0 & -\lambda_{31} & 0 & 1 \end{array} \right) \left(\begin{array}{c} \tilde{a}_0^T \\ \tilde{a}_1^T \\ \tilde{a}_2^T \\ \tilde{a}_3^T \end{array} \right) = \left(\begin{array}{c} \tilde{a}_0^T \\ \tilde{a}_1^T \\ \left(\tilde{a}_2^T \right) - \left(\lambda_{21} \right) \tilde{a}_1^T \\ \left(\tilde{a}_3^T \right) - \left(\lambda_{31} \right) \tilde{a}_1^T \end{array} \right) = \left(\begin{array}{c} \tilde{a}_0^T \\ \tilde{a}_1^T \\ \tilde{a}_2^T - \lambda_{21} \tilde{a}_1^T \\ \tilde{a}_3^T - \lambda_{31} \tilde{a}_1^T \end{array} \right).$$

Notice the similarity with what one does in Gaussian elimination: take multiples of one row and subtracting these from other rows.

Homework 5.2.4.1. Evaluate

$$\left(\begin{array}{c|ccc} I_k & 0 & 0 & 0 \\ \hline 0 & 1 & 0 & 0 \\ 0 & -l_{21} & I & \end{array} \right) \left(\begin{array}{c|cc} A_{00} & a_{01} & A_{02} \\ \hline 0 & \alpha_{11} & a_{12}^T \\ 0 & a_{21} & A_{22} \end{array} \right)$$

where I_k is the $k \times k$ identity matrix and A_0 has k rows. If we compute

$$\left(\begin{array}{c|ccc} A_{00} & a_{01} & A_{02} \\ \hline 0 & \alpha_{11} & a_{12}^T \\ 0 & \hat{a}_{21} & \hat{A}_{22} \end{array} \right) := \left(\begin{array}{c|ccc} I_k & 0 & 0 & 0 \\ \hline 0 & 1 & 0 & 0 \\ 0 & 0 & -l_{21} & I \end{array} \right) \left(\begin{array}{c|cc} A_{00} & a_{01} & A_{02} \\ \hline 0 & \alpha_{11} & a_{12}^T \\ 0 & a_{21} & A_{22} \end{array} \right)$$

how should l_{21} be chosen if we want $\hat{a}_{21}$ to be a zero vector?

▼ [Solution](#)

$$\begin{aligned} & \left(\begin{array}{c|ccc} I_k & 0 & 0 & 0 \\ \hline 0 & 1 & 0 & 0 \\ 0 & -l_{21} & I & \end{array} \right) \left(\begin{array}{c|cc} A_{00} & a_{01} & A_{02} \\ \hline 0 & \alpha_{11} & a_{12}^T \\ 0 & a_{21} & A_{22} \end{array} \right) \\ &= \left(\begin{array}{c|ccc} A_{00} & a_{01} & A_{02} \\ \hline 0 & \alpha_{11} & a_{12}^T \\ 0 & -l_{21}\alpha_{11} + a_{21} & -l_{21}a_{12}^T + A_{22} \end{array} \right) \\ &= \left(\begin{array}{c|ccc} A_{00} & a_{01} & A_{02} \\ \hline 0 & \alpha_{11} & a_{12}^T \\ 0 & a_{21} - \alpha_{11}l_{21} & A_{22} - l_{21}a_{12}^T \end{array} \right) \end{aligned}$$

If $l_{21} = a_{21}/\alpha_{11}$ then $\tilde{a}_{21} = a_{21} - \alpha_{11}a_{21}/\alpha_{11} = 0$.

Hopefully you notice the parallels between the computation in the last homework, and the algorithm in [Figure 5.2.1.1](#).

Now, assume that the right-looking LU factorization has proceeded to where A contains

$$\left(\begin{array}{c|cc} A_{00} & a_{01} & A_{02} \\ \hline 0 & \alpha_{11} & a_{12}^T \\ 0 & a_{21} & A_{22} \end{array} \right),$$

where A_{00} is upper triangular (recall: it is being overwritten by U !). What we would like to do is eliminate the elements in a_{21} by taking multiples of the "current row" $\left(\begin{array}{c|cc} \alpha_{11} & a_{12}^T \end{array} \right)$ and subtract these from the rest of the rows:

$\left(\begin{array}{c|cc} a_{21} & A_{22} \end{array} \right)$ in order to introduce zeroes below α_{11} . The vehicle is an appropriately chosen Gauss transform, inspired by [Homework 5.2.4.1](#). We must determine l_{21} so that

$$\left(\begin{array}{c|cc} I & 0 & 0 \\ \hline 0 & 1 & 0 \\ 0 & -l_{21} & I \end{array} \right) \left(\begin{array}{c|cc} A_{00} & a_{01} & A_{02} \\ \hline 0 & \alpha_{11} & a_{12}^T \\ 0 & a_{21} & A_{22} \end{array} \right) = \left(\begin{array}{c|cc} A_{00} & a_{01} & A_{02} \\ \hline 0 & \alpha_{11} & a_{12}^T \\ 0 & 0 & A_{22} - l_{21}a_{12}^T \end{array} \right).$$

As we saw in [Homework 5.2.4.1](#), this means we must pick $l_{21} = a_{21}/\alpha_{11}$. The resulting algorithm is summarized in [Figure 5.2.4.3](#). Notice that this algorithm is, once again, identical to the algorithm in [Figure 5.2.1.1](#) (except that it does not overwrite the lower triangular matrix).

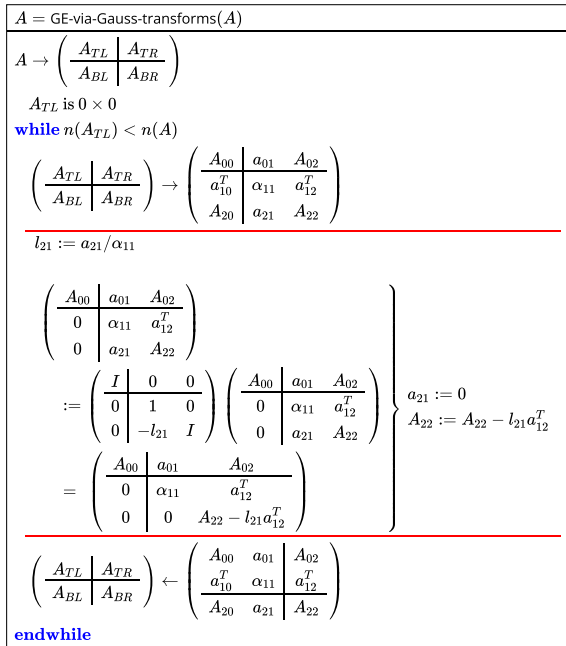


Figure 5.2.4.3. Gaussian elimination, formulated as a sequence of applications of Gauss transforms.

Homework 5.2.4.2. Show that

$$\left(\begin{array}{c|cc} I_k & 0 & 0 \\ \hline 0 & 1 & 0 \\ 0 & -l_{21} & I \end{array} \right)^{-1} = \left(\begin{array}{c|cc} I_k & 0 & 0 \\ \hline 0 & 1 & 0 \\ 0 & l_{21} & I \end{array} \right)$$

where I_k denotes the $k \times k$ identity matrix.

▼ [Hint](#) ▼ [Solution](#)

$$\begin{aligned} & \left(\begin{array}{c|cc} I_k & 0 & 0 \\ \hline 0 & 1 & 0 \\ 0 & -l_{21} & I_{(n-k-1) \times (n-k-1)} \end{array} \right) \left(\begin{array}{c|cc} I_k & 0 & 0 \\ \hline 0 & 1 & 0 \\ 0 & l_{21} & I \end{array} \right) \\ &= \left(\begin{array}{c|cc} I_k & 0 & 0 \\ \hline 0 & 1 & 0 \\ 0 & -l_{21} + l_{21} & I \end{array} \right) \\ &= \left(\begin{array}{c|cc} I_k & 0 & 0 \\ \hline 0 & 1 & 0 \\ 0 & 0 & I \end{array} \right) \end{aligned}$$

To show that $B = A^{-1}$, it suffices to show that $BA = I$ (if A and B are square).

Starting with an $m \times m$ matrix A , the algorithm computes a sequence of m Gauss transforms $L_0, \dots, L_{m-1}$, each of the form

$$L_k = \left(\begin{array}{c|cc} I_k & 0 & 0 \\ \hline 0 & 1 & 0 \\ 0 & -l_{21} & I \end{array} \right), \quad (5.2.2)$$

such that $L_{m-1}L_{m-2} \cdots L_1L_0A = U$. Equivalently, $A = L_0^{-1}L_1^{-1} \cdots L_{m-2}^{-1}L_{m-1}^{-1}U$, where

$$L_k^{-1} = \left(\begin{array}{c|cc} I_k & 0 & 0 \\ \hline 0 & 1 & 0 \\ 0 & l_{21} & I \end{array} \right).$$

It is easy to show that the product of unit lower triangular matrices is itself unit lower triangular. Hence

$$L = L_0^{-1} L_1^{-1} \dots L_{n-2}^{-1} L_{n-1}^{-1}$$

is unit lower triangular. However, it turns out that this L is particularly easy to compute, as the following homework suggests.

Homework 5.2.4.3. Let

$$\tilde{L}_{k-1} = L_0^{-1} L_1^{-1} \dots L_{k-1}^{-1} = \left(\begin{array}{c|cc} L_{00} & 0 & 0 \\ \hline l_{10}^T & 1 & 0 \\ L_{20} & 0 & I \end{array} \right) \quad \text{and} \quad L_k^{-1} = \left(\begin{array}{c|cc} I_k & 0 & 0 \\ \hline 0 & 1 & 0 \\ 0 & l_{21} & I \end{array} \right),$$

where L_{00} is a $k \times k$ unit lower triangular matrix. Show that

$$\tilde{L}_k = \tilde{L}_{k-1} L_k^{-1} = \left(\begin{array}{c|cc} L_{00} & 0 & 0 \\ \hline l_{10}^T & 1 & 0 \\ L_{20} & l_{21} & I \end{array} \right).$$

▼ [Solution](#)

$$\begin{aligned} \tilde{L}_k &= L_0^{-1} L_1^{-1} \dots L_{k-1}^{-1} L_k^{-1} = \tilde{L}_{k-1} L_k^{-1} \\ &= \left(\begin{array}{c|cc} L_{00} & 0 & 0 \\ \hline l_{10}^T & 1 & 0 \\ L_{20} & 0 & I \end{array} \right) \left(\begin{array}{c|cc} I_k & 0 & 0 \\ \hline 0 & 1 & 0 \\ 0 & l_{21} & I \end{array} \right) \\ &= \left(\begin{array}{c|cc} L_{00} & 0 & 0 \\ \hline l_{10}^T & 1 & 0 \\ L_{20} & l_{21} & I \end{array} \right). \end{aligned}$$

What this exercise shows is that $L = L_0^{-1} L_1^{-1} \dots L_{n-2}^{-1} L_{n-1}^{-1}$ is the triangular matrix that is created by simply placing the computed vectors l_{21} below the diagonal of a unit lower triangular matrix. This insight explains the "magic" observed in [Homework 5.2.1.3](#). We conclude that the algorithm in [Figure 5.2.1.1](#) overwrites $n \times n$ matrix A with unit lower triangular matrix L and upper triangular matrix U such that $A = LU$. This is known as the LU factorization or LU decomposition of A .

Ponder This 5.2.4.4. Let

$$L_k = \left(\begin{array}{c|cc} I_{k \times k} & 0 & 0 \\ \hline 0 & 1 & 0 \\ 0 & -l_{21} & I \end{array} \right).$$

Show that

$$\kappa_2(L_k) \geq \|l_{21}\|_2^2.$$

What does this mean about how error in A may be amplified if the pivot (the α_{11} by which entries in a_{21} are divided to compute l_{21}) encountered in the right-looking LU factorization algorithm is small in magnitude relative to the elements below it? How can we choose which row to swap so as to minimize $\|l_{21}\|_2$?

▼ [Hint](#)

Revisit [Homework 1.3.5.5](#).